

Todd Takala

Lead Data Scientist, Machine Learning Engineer, AI/ML and Analytics

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Summary

Data Scientist & Machine Learning Engineer with strong leadership in machine learning and statistical modeling who generated over \$1.3 billion in business value. Specializations include design of data solutions, implementation of MLOps pipelines on AWS and Snowflake, enhancing customer engagement.

Professional Experience

Lead Data Scientist - Caterpillar Inc. - Surprise, AZ

09/2022 to Present

- Architected and developed end-to-end data products using programming skills in Python and R, including an award-winning AI platform to predict equipment repair times, resulting in an \$80 million reduction in warranty costs and a CEO Award
- Collaborated with senior leadership and stakeholders to facilitate understanding of the project's broad impact and create precise user requirements, ensuring alignment with the business road map through written and verbal communication
- Used LLMs and natural language processing to find similarities in service jobs and performed feature engineering to retain important parameters, which increased model accuracy by 40%
- Conducted exploratory data analysis on the training datasets, eliminating outliers using sophisticated analytics such as isolation forest and anomaly detection techniques
- Trained AI models using Python and SageMaker, applying machine learning algorithms like neural networks, PyTorch, XGBoost, TensorFlow, and LightGBM, which improved model performance and enhanced the accuracy of equipment repair time predictions, leading to better maintenance scheduling
- Engineered production-ready MLOps pipelines on AWS and Snowflake, establishing best practices for cloud architecture. Managed project lifecycles, automated enterprise networks, and implemented monitoring strategies to assess model performance and business outcomes

Reliability Analytics Manager - Amazon.com - Tempe, AZ

02/2022 to 09/2022

- Applied advanced analytics, mathematics, and statistics to create a new KPI for autonomous robots used in warehouse automation through big data analysis, and wrote a white paper on statistical programming, resulting in its adoption by senior management
- Led a team of business intelligence engineers and data analysts to deploy a pipeline using Shiny R, which improved real-time communication of robot status at all global fulfillment centers, enhancing operational efficiency and data analytics capabilities
- Designed and implemented solutions utilizing large databases with Redshift and PostgreSQL, supporting data-driven decision-making and enhancing business intelligence capabilities

Data Engineering Manager - Empire-Cat - Mesa, AZ

01/2021 to 02/2022

- Developed time-series AI model and MLOps pipeline for demand forecasting resulting in the identification of key supply chain risks.
- Pivotal leader who guided a team in designing, building, and maintaining the robust and scalable data infrastructure that formed the backbone of Empire's data-driven organization
- Created predictive ML model for engine reliability, durability, repair, preventive maintenance and intervention leading to \$62 million in savings within 90 days

Reliability Engineering Manager - Empire-Cat - Mesa, AZ

07/2016 to 01/2021

- Supervised 8 engineers and technical communicators and acted as a customer-facing expert on mining equipment. Mentored up to 6 interns per year
- Headed up the root cause failure analysis program for Empire by taking a data, facts, and evidence approach, which fed the FMEA and 6 Sigma process at Caterpillar
- Implemented a data-driven failure analysis program, boosting an aging fleet's physical availability to over 90%
- Optimized resource utilization and reduced costs by leading successful reuse and salvage decisions of machinery parts

Lead Reliability Engineer - Empire-Cat - Mesa, AZ

01/2016 to 07/2016

- Led initiatives to maintain and expand corporate relationships with mining clients.
- Managed testing and experiment program and provided customer insights to Caterpillar engineering teams.
- Focused on deploying large off-highway truck design improvements to enhance operational efficiency.

Technical Communications Manager - Empire-Cat - Mesa, AZ

12/2012 to 01/2016

- Applied the DMAIC Six Sigma methodology and computer science with complex problem-solving to identify opportunities for improving operation, manufacturing, and maintenance of equipment fleet, yielding \$1.2 billion in savings over 6 years
- Led a team of 5 Technical Communicators in a client-facing role to effectively convey customer feedback to Caterpillar's engineering organizations, driving improvements in product development and customer satisfaction
- Built trust and enhanced customer experience and relationships, promoting data-based decisions that improved business outcomes
- Partnered on design of experiments to improve reliability, safety and durability
- Developed Python and MS-SQL application to track performance and outcomes of prototype designs tested in the field

Reliability Engineer - Empire-Cat - Mesa, AZ

01/2012 to 12/2012

- Created statistical oil analysis application to analyze heavy equipment fleets according to ASTM D 7720, which became ground truth for Empire condition monitoring program
- Provided expertise in mechanical engineering and heavy equipment, serving as a key resource for improving complex machinery systems
- Investigated the primary causes of machinery downtime at mining locations, identifying key issues and recommending solutions to improve operational efficiency

Certification & Technical Registration

- **Arizona State Board of Technical Registration:** EIT, Mechanical Engineering, Registration No. 10670
- **Canonical:** Ubuntu Linux Professional Certificate
- **Data Camp:** AI Fundamentals Certificate • Certified SQL Associate, Data Scientist Professional Certificate
- **GitHub:** Career Essentials in GitHub Professional Certificate
- **IBM:** Advanced Deep Learning Specialist • RAG and Agentic AI Professional Certificate • Deep Learning with PyTorch, Keras and Tensorflow Professional Certificate
- **Komatsu North America:** Certified Technical Communicator
- **US Department of Labor:** Operating Engineer Vocational Certificate

Education

Bachelor of Science in Mechanical Engineering - Arizona State University - Tempe, AZ

06/2003 to 05/2006

Hard Skills

A/B Testing, Agentic AI, Agile Methodology, AI/ML, Anomaly Detection, Amazon Bedrock, Artificial Intelligence, AWS, BASH, CI/CD, Computer Science, RDBMS, Database Administrator, Data Science, DBT, Deep Learning, Design of Experiments, DevOps, Docker, Embedding, ETL, Fine Tuning, Advanced Excel, Financial Forecasting, Generative AI, Git, Linear Algebra, Large Language Models, LLM, Mathematics, Natural Language Processing, NLP, Pandas, Power BI, Public Cloud, Python, PyTorch, R, RAG, Research, Sagemaker, Scikit-learn, Sagemaker, Scripting Language, SDLC, Simulation, SnapLogic, Snowflake, SQL Queries, Statistical Analysis, Streamlit, SQL, Tableau, TensorFlow, Transact SQL, Transformers, Version Control

Soft Skills

Adaptability, Attention to Detail, Business Requirements, Coaching, Communication Skills, Continuous Improvement, Creative Problem Solving, Data Driven, Decision making, Growth Mindset, Interviewing, Leadership Experience, Life Long Learning, Listening, Mentoring, Methodical, Multicultural Collaboration, Open minded, Problem Solving, Professional Services, Presentations, Public Speaking, Strategic Thinking, Team Building, Thought Leadership, Willing to Learn

Courses

- **Amazon Machine Learning University:** Tabular Data with Autogluon
- **ASME:** Y14.5M Geometric Dimensioning and Tolerancing
- **Caterpillar:** Applied Failure Analysis • SQL Server Toolset & Services • Business Tools for Service Managers • Improving Component Durability
- **Data Camp:** Bayesian Modeling in RJAGS
- **DDI:** Communicating for Leadership Success • Coaching for Peak Performance • Delegating with Purpose • Setting Performance Expectations • Resolving Workplace Conflict
- **DeepLearning.AI:** Building Systems with the ChatGPT API • ChatGPT Prompt Engineering for Developers • Finetuning Large Language Models
- **Franklin Covey:** Leadership: Great leaders - Great Teams - Great Results
- **Harper College:** Pneumatics and Hydraulics
- **IBM:** Advanced RAG with Vector Databases and Retrievers • Agentic AI with LangChain and LangGraph • Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI • Build RAG Applications • Deep Learning with Keras and Tensorflow • Deep Learning & Neural Networks with Keras • Vector Databases for RAG • Deep Learning with PyTorch
- **LinkedIn Learning:** Advanced Snowflake: Deep Dive Cloud Data Warehousing and Analytics • Building Applications Using Amazon Bedrock • Data Engineering with dbt • Designing Machine Learning Systems in the Cloud • Learning Kubernetes
- **Noria:** Oil Analysis I, II, III
- **O'Reilly Media:** Designing Machine Learning Systems in the Cloud
- **SAE:** Principles of Cost and Finance for Engineers